

FAR WESTERN UNIVERSITY
Semester End Examination-2079
English Grammar and Composition (CSIT.111)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: First

Full Marks: 100

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – 'A'

Attempt all questions (Very Short Questions)

8×3=24

14

1. Why do you think that people are more interested in themselves than in other people?
2. Why should supertasters and nontasters pay close attention to the foods they eat?
3. Do you think sponsoring games is a good or bad investment? Explain.
4. What is bystander effect?
5. Has an advertisement ever helped you in some way? Explain how it helped.
6. Why do you think we perceive activities such as mountain climbing to be riskier than they actually are?
7. Why will there be no cars in Masdar?
8. What kind of organizations typically ask for donation? Why?

Group – 'B'

Attempt any five questions. (Short Questions)

5×8=40

25

9. Are there any topics of conversation you should avoid when meeting someone for the first time? What are they? Why should we avoid them?
10. Do you think you are a nontaster, medium taster, or supertaster? Why?
11. What are some things people give up or sacrifice in order to be successful?
12. Would you consider joining a website like Second Life? Why or why not?
13. What factors might make someone choose not to help a stranger?
14. Do you think that people who do extreme sports would not be happy if they didn't continue to take risks? In other words, do you think they need to take risks?

P.T.O.

Group – 'C'

24

Attempt any three questions. (Long Questions)

3×12=36

15. Can you think of some advantages to living in a city like Masdar? Can you think of disadvantages?
16. How Beatrice's family was supported and how Beatrice was able to achieve her goals after she received a goat as a help from the NGO?
17. What are the advantages and disadvantages of using a social networking site?
18. Do you agree that showing an interest in other people is the best way to make a good first impression? What other ways are there to make a good first impression?

W

$$\begin{array}{r} 28 + 24 + 14 \\ \hline 50 + 13 \\ \hline = 63 \end{array}$$

FAR WESTERN UNIVERSITY

Semester End Examination-2079

Information Technology Fundamentals (CSIT.112)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Full Marks: 100

Semester: First

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – 'A'

Attempt all questions (Very Short Questions)

8×3=24

1. Why computer is called versatile machine?
2. What do you mean by mobile computing?
3. Convert $(A5B)_{16}$ into decimal number.
4. State and prove De-Morgan's theorem.
5. What is cache memory?
6. What are the methods of file organization?
7. Differentiate between compiler and interpreter.
8. Explain about BCR.

Group – 'B'

Attempt any five questions. (Short Questions)

5×8=40

9. What is software? Differentiate between system software and application software.
10. What are the similarities and dissimilarities between IBM PC and IBM compatible? Explain.
11. Subtract $(110110)_2$ from $(1101)_2$ by using 1's and 2's complement method.
12. Differentiate between impact and non-impact printers.
13. What are the types of programming error? How are they detected?
14. What is search engine? List any five examples of search engine.

P.T.O.

Group – 'C'

Attempt any three questions. (Long Questions)

3×12=36

15. Explain the significance of logic gate. Describe different logic gates along with their logical symbols and truth table.
16. Define computer architecture. Explain the components of computer system with block diagram.
- ✓ 17. What is operating system? Explain different functions of operating system.
18. Define network topology. Explain different types of topologies with their merits and demerits.

FAR WESTERN UNIVERSITY
Semester End Examination-2079

Calculus and Analytical Geometry (CSIT.113)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: First

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Group – 'A'

Attempt all questions (Very Short Questions)

8×3=24

1. Define the bounded and unbounded sequences with example. Identify whether the sequence $\langle u_n \rangle = \frac{n}{n+1}$ is bounded or not.
2. Define limit superior and limit inferior of a sequence $\langle u_n \rangle$ with examples. Find the limit superior and limit inferior of $\langle u_n \rangle = (-1)^n (2 + \frac{1}{n})$, $n \in \mathbb{N}$
3. Define convergent, divergent and oscillating series with example. Test whether the series $\sum (-1)^n$ is convergent, divergent or oscillating.
4. What do you mean by n^{th} derivative of any function $y = f(x)$. Find n^{th} derivative of $y = \sin x$
5. What is Beta and gamma function. Show that $\Gamma(n+1) = n \Gamma n$
6. Define eccentricity and classify the different conics by eccentricity and state the condition that when eccentricity tends to zero.
7. Discuss the motion of a particle whose position $P(x, y)$ at time t is given by $x = a \cos t$, $y = -b \sin t$, $0 \leq t \leq 2\pi$
8. Find area bounded by lines $y=0$, $y=x^2$ and $x=1$. Using double integral.

Group – 'B'

Attempt any five questions. (Short Questions)

5×8=40

9. Prove the series $\sum r^n$ converges to $\frac{r}{1-r}$ if $|r| < 1$. It diverges to $+\infty$ if $r \geq 1$. It oscillates finitely for $r=-1$. Also find the sum of series $\sum \frac{1}{2^n}$.
10. State Leibnitz theorem for n^{th} derivative. Using it if $y = (x^2-1)^n$, prove that $(x^2-1)y_{n+2} + 2xy_{n+1} - n(n+1)y_n = 0$. (2+6=8)
11. Obtain the reduction formula for $\int_0^{\pi/2} \sin^n x \, dx$ and hence evaluate $\int_0^{\pi/2} \sin^7 x \, dx$. (6+2=8)

12. Discuss the meaning of $\int_{\theta=\alpha}^{\theta=\beta} f(\theta) d\theta$. Using the concept of integration in polar co-ordinate find area bounded by the polar curve $r = a(1 - \cos\theta)$. (3+5=8)
13. Define TNB system of moving particle along any smooth curve in space. If the space curve is given by $\vec{r}(t) = 5\cos t \vec{i} + (1+3\sin t) \vec{j} + 4\sin t \vec{k}$, find $T(t)$, $N(t)$, $B(t)$ (2+6=8)
14. Define meaning of $\iint_R f(x, y) dx dy$. Evaluate $\int_0^5 \int_0^{x^2} x(x^2 + y^2) dx dy$, by reversing the order of integration. (8)

Group – 'C'

Attempt any three questions. (Long Questions)

3×12=36

15. Discuss the different techniques of testing the convergence of an infinite series of positive terms with example. Using appropriate method test the convergence of an infinite series $\sum \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2 \cdot 4 \cdot 6 \dots 2n} x^n$, $x > 0$. (4+8=12)
16. Discuss the concept of limit, continuity and derivability of multivariable function with suitable examples. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{2xy}{x^2+3y^2}$ (i) along path $y=x^2$ (ii) along $y=3x$ (iii) along x-axis and along y-axis. Does function is continuous at (0,0)? If not why? (4+6+2=12)
17. Discuss the meaning of extrema of two and three variable function. Find the extreme value of the function $f(x, y) = 2(x-y)^2 - x^4 - y^4$. (4+8=12)
18. (a) If $u = \log(x^3 + y^3 + z^3 - 3xyz)$ show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$ (6)
- (b) Discuss the improper integral and its types with examples. Test the convergence of $\int_1^{\infty} \frac{1}{(3x+1)^2} dx$ (6)

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FAR WESTERN UNIVERSITY
Semester End Examination-2079
Electronic Principles (CSIT.114)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: First

Full Marks: 100

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – 'A'

Attempt all questions (Very Short Questions)

8×3=24

1. Define voltage source and current source.
2. What is Barkhausen criterion for oscillation?
3. Define pinch-off voltage and gate-source cut-off voltage.
4. Differentiate between BJT and FET.
5. Explain the significance of Q-point in a dc load line.
6. What happens to the width of the depletion layer when reverse biasing is increased across the junction of p-n junction diode?
7. Define regulated and unregulated power supply.
8. Describe in brief about virtual ground concept.

Group – 'B'

Attempt any five questions. (Short Questions)

5×8=40

9. What do you understand by Thevenin's theorem? Draw a suitable circuit of your own to explain Thevenin's theorem.
10. Define operational amplifier. Explain the working of an inverting operational amplifier and derive an expression for closed loop gain.
11. Discuss about the common base mode characteristic of a NPN transistor with suitable diagram.
12. Define photodiode? Discuss about its construction and theory.
13. Why biasing of a transistor is important? How does transistor act as a switch, explain?
14. Define oscillator? With well label diagram discuss about Hartely oscillator.

P.T.O.

$$I = \left(\frac{E}{R_1 + R_2} \right)$$

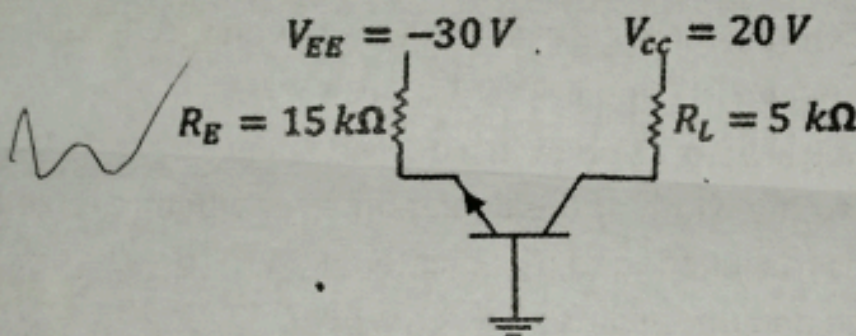
$$V_3 = \frac{I R_3}{\left(\frac{R_3}{R_1 + R_2} \right) E}$$

Group - 'C'

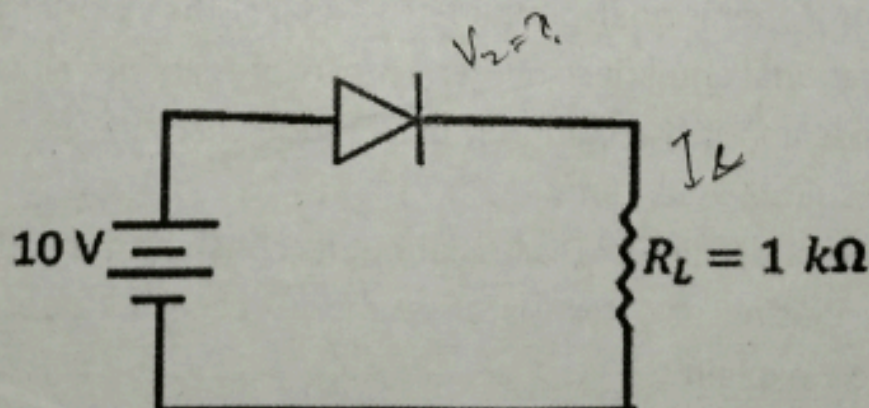
Attempt any three questions. (Long Questions)

3×12=36

15. (a) What are the difference between Zener diode and p-n junction diode? How does Zener diode regulate the voltage? Explain with necessary theory. (8)
- (b) A transistor has $\alpha = 0.98$, $I_B = 100 \mu A$ and $I_{CO} = 6 \mu A$. Calculate I_C and I_E . (4)
16. (a) Draw a circuit diagram for the common collector amplifier and obtain relation for its voltage gain. Explain why there is phase inversion between input and output? (8)
- (b) For the given circuit diagram draw the dc load line. (4)



17. (a) Define MOSFET? Explain the working of enhancement mode MOSFET and draw its static characteristics. (8)
- (b) Show that for adder circuit output voltage is proportional to the algebraic sum of two or more input voltages. (4)
18. (a) Define rectifier. Discuss about full wave rectifier. (8)
- (b) Use the second approximation to calculate the load voltage, load current and diode power in given figure. (4)



FAR WESTERN UNIVERSITY

Semester End Examination-2079

Programming Fundamentals and 'C' Programming (CSIT.115)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Full Marks: 100

Semester: First

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group – 'A'

Attempt all questions (Very Short Questions)

8×3=24

1. Why programming tools are necessary to develop program?
2. What is variable? How it is declared?
3. Differentiate between print f () and puts () function.
4. What is FILE pointer?
5. What are actual and formal parameters?
6. What is entry controlled loop? Give example.
7. Can you use continue statement without using loop? Give reasons.
8. Describe the header file and its usages in 'C' programming with examples.

Group – 'B'

Attempt any five questions. (Short Questions)

5×8=40

9. What is flow chart? Describe symbols used in flow charts. Write flowchart to find largest number among three numbers?
10. Briefly describe the structure of 'C' program.
11. What is pointer? Write advantages and disadvantages of pointers in 'C'? Give programming examples of pointer.
12. Write 'C' program to display prime numbers between 100 to 400.
13. Describe various file opening modes. Write 'C' program that read data of student from input device and write that data on "student. dat" file.
14. Write 'C' program to enter a string and display its reverse without using string manipulation function.

P.T.O.

Group – 'C'

Attempt any three questions. (Long Questions)

3×12=36

15. What is user defined function? Describe various types of user defined function with programming examples.
16. Write 'C' program to display multiplication of two matrices.
17. Write 'C' program to read 'N' students record (roll no, Name, faculty, DOB (dd, mm, yy)) by using structure and display the record of students who belongs to "CSIT" faculty.
18. Describe various types of string manipulations function?

